JANGARA BLISS

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EDUCATION

Fort Lewis College (FLC), Durango, CO | B.S. Computer Engineering Expected May 2027
Minors in Mathematics and Business Administration | GPA 3.65 cumulative, 3.80 upper-division, 4.00 in Spring 2026 (17 cr.)
Coursework: Intro to Robotics; Systems & Control; CE 433 Embedded Devices (FPGA design); Digital Signal Processing; Engineering Statistics (calculus-based); Differential Equations; Calculus III. Completing before enrollment: Discrete Math, Analog Electronics, Linear Algebra, CE 432 Practical Robot Design, Semiconductor Fab (FLC nanolab).

RESEARCH & ENGINEERING EXPERIENCE

Research Assistant, Humanoid Robot Intelligence | Advisor: Dr. Yiyan Li | Booster K1, solo project May 2026 – Aug 2026

- Built a two-tier stack: NaVILA, an 8B-parameter vision-language navigation model (~1 Hz), commanding a reinforcement-learning locomotion policy I trained in Isaac Sim/Isaac Lab (50 Hz; works in simulation only).
- Deployed NaVILA live on the physical K1 for two natural-language episodes with zero teleoperation, via a relay I built (RTX 5090 VLM inference, ~350 ms; Booster SDK velocity control; Booster's built-in walker).
- Integrated the K1 into VLN-CE-Isaac and ran all 1,077 episodes; pipeline fixes and policy retraining raised success from under 1% to 18.3% (22 training runs, ~270 training GPU-h). Per-episode records and analysis scripts are public: github.com/Janga786/k1-navila-research.
- Ran a preregistered 2,400-episode ablation; two identically configured 300-episode runs agreed in aggregate (15.0% vs. 16.0%) yet disagreed on 11.0% of episodes (kappa 0.58).
- Built a ROS 2 cart-seek stack (YOLO-World): K1 walks to a cart, stops ~1 m away, circles it to read a QR code; ~25 live runs.

Senior Design: Bridge Crack-Inspection Arm | Advisor: Dr. Yiyan Li | reBot B601-DM, solo build Sep 2026 – present

- Brought up all seven motors (6+1 DoF) over CAN with a 200 Hz control loop (setpoint, stiffness, damping, feed-forward torque per motor); built the Jacobian, inverse kinematics, and a minimum-jerk planner, rehearsing new motion scripts in simulation before hardware runs.
- Measured the elbow's gravity load with a slow torque test (6.6 N·m vs. 2.7 N·m in the URDF); gravity/friction feed-forward and gain tuning cut loaded-joint holding error from 0.04–0.05 rad to at most 0.005 rad (41 logged runs).

Research Assistant, Computer Vision & Robotics | Advisor: Dr. Kevin Wedeward | Solo project May 2025 – Sep 2025

- Revived two non-functional Rethink robots (Sawyer, Baxter) with no vendor support: traced Sawyer's boot failure to a dead CMOS battery that had reset its BIOS into an incompatible Secure Boot setting, obtained the Intera update from other universities, and set up ROS 1/ROS 2 with Docker, MoveIt, and Gazebo.
- Trained a YOLOv8 thermal hotspot detector on Blender synthetic data; with Baxter holding the camera, it flagged an induced hotspot on test panels in every saved frame. 2nd Place, FLC Physics & Engineering Symposium (top solo project).

Team Lead, Colorado Robotics Challenge (NASA Space Grant) | Advisor: Dr. Charles Hakes Oct 2025 – Apr 2026

- Led a 4-person team; designed the electrical system of an 18-DoF hexapod (Arduino Mega, 9-DoF IMU, bump sensors, LiPo + SBEC power) and extended its firmware: gyro-magnetometer heading hold, gyro-bias calibration, bump avoidance.
- Later built a Kuramoto coupled-oscillator gait simulator (tripod, wave, ripple from one system); open-sourced, 34-test CI.

Research Assistant, AI & Data | Advisor: Dr. Matthew Welz | Solo project Aug 2024 – Oct 2025

- Built the FLC radio station's music-library database (Power Apps; designed for 60+ daily users) and prototyped embedding-based artist matching and LLM agents over Neo4j; the station later deployed the database on another platform.

AWARDS, HONORS & PROGRAMS

2nd Place, FLC Physics & Engineering Symposium (2025) | New Registered Student Organization of the Year (2023–24)
Katz School of Business Leadership Award (2024) | Goldman Sachs Emerging Leaders Series (2024) | Dean's List (2026)

LEADERSHIP & SERVICE

- **Co-Founder & President, Entrepreneurial Ventures Association** (2023–24): led an 8-person executive team; ran a $1,500 micro-grant pitch competition; helped bring NASA's Technology Transfer University (T2U) program to campus.
- **Student Representative, FLC Strategic Implementation Committee** (2025–26): one of four students working with the college president, trustees, and deans on the 2025–2030 Strategic Plan.
- **STEM Tutor** (2024–25) | **Chip Competition Manager**, Nicholas Cisewski Memorial Golf Tournament for pediatric cancer research (2020–present)

SKILLS

Control: joint PD + gravity/friction feed-forward (200 Hz over CAN), Jacobian and damped-least-squares IK, minimum-jerk trajectories, IMU-fused heading hold, Kuramoto CPG gaits, RL locomotion (PPO, Isaac Lab) | **Robotics & ML:** PyTorch, ROS 1/ROS 2, MoveIt, Isaac Lab, MuJoCo, Gazebo, NaVILA (VLN) deployment, YOLOv8/YOLO-World, Blender synthetic data, Open3D, LLM agents on Neo4j, AI coding agents & orchestration (Claude Code, Codex, local LLMs) | **Hardware:** board bring-up, soldering, embedded C drivers (GPIO, PWM, ADC, UART), ATmega PCB design, CAN bus, oscilloscope, logic analyzer, FPGA (Verilog, Vivado) | **Programming:** Python, C++, C, Verilog, Cypher; Linux, Docker, Git | Spanish (conversational)